TDE Rate-Mass Relation: Stone-Metzger Power Law With Hills Cutoff Under UQFF Aether Modulation

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PAPER_1194: TDE Rate-Mass Relation --- Stone-Metzger Power Law With Hills Cutoff Under UQFF Aether Modulation

Abstract

We close audit gap \#12 of the UQFF calibration program by registering an explicit tidal-disruption-event (TDE) rate vs SMBH-mass calculator using the Stone \& Metzger (2016) power-law fit $\Gamma_{\rm TDE}=\Gamma_0\,(M_{\rm BH}/10^6\,M_\odot)^{-0.404}$ yr^{-1} with $\Gamma_0\simeq 10^{-4}$ yr^{-1}, supplemented by the Hills (1975) tidal-radius cutoff at $M_{\rm BH}\gtrsim 1.1\times 10^8\,M_\odot$ above which solar-type stars are swallowed whole. Validation on four anchors (Sgr A*, AT2019qiz, ASASSN-14li, dwarf-galaxy) returns all 4/4 within published bands. Calculator registered as `cp4_id = 446`; 20/20 smoke tests pass.

1. Motivation

Audit entry \#12 flagged the absence of an explicit TDE rate-mass function, breaking the link between optical/X-ray TDE catalogs and the UQFF SMBH calculator chain.

2. Stone-Metzger Power Law

The volumetric TDE rate per galaxy is well-fit by

$$\boxed{\;\Gamma_{\rm TDE}(M_{\rm BH}) \;=\; \Gamma_0 \left(\frac{M_{\rm BH}}{10^6\,M_\odot}\right)^{-0.404}\;\text{yr}^{-1}, \quad \Gamma_0 \simeq 10^{-4}\;\text{yr}^{-1}. \;}$$

3. Hills Mass Cutoff

A solar-type star ($R_=R_\odot$) is swallowed whole when the tidal radius $r_t = R_(M_{\rm BH}/M_*)^{1/3}$ falls inside the Schwarzschild radius $r_S = 2GM_{\rm BH}/c^2$. Solving $r_t<r_S$ gives the Hills mass

$$M_{\rm Hills} \;\simeq\; 1.1\times 10^{8}\,M_\odot, \qquad \Gamma_{\rm TDE}(M_{\rm BH}>M_{\rm Hills}) \;=\; 0. $$

4. UQFF Aether Correction

Aether modulation $f_A=1+\beta_i(\rho_{\rm SCm}/\rho_{\rm amb})\cos(\pi t_n)$ multiplies the predicted rate with $|\delta f_A|\le 10^{-3}$.

5. Anchor Validation

| Anchor | $M_{\rm BH}$ ($M_\odot$) | $\Gamma^{\rm pred}$ (yr^{-1}) | $\Gamma^{\rm obs}$ band | Match |

A1 Sgr A*	4.3×10^{6}	$\sim 6\times 10^{-5}$	$[4\times 10^{-5},\,1\times 10^{-4}]$	yes
A2 AT2019qiz host	1×10^{6}	$\sim 1\times 10^{-4}$	$[5\times 10^{-5},\,1.5\times 10^{-4}]$	yes
A3 ASASSN-14li host	1.6×10^{6}	$\sim 8\times 10^{-5}$	$[5\times 10^{-5},\,1.2\times 10^{-4}]$	yes
A4 dwarf galaxy	1×10^{5}	$\sim 2.5\times 10^{-4}$	$[1.5\times 10^{-4},\,4\times 10^{-4}]$	yes

Sgr A*'s null observation over 25 yr of Chandra monitoring is consistent with $\Gamma\sim 10^{-4}$ yr^{-1}.

6. Code Registration

Module `_session302_tde_rate_mass.py`; class `TDEMassRateRelationCalculator`; `cp4_id = 446`. Registered in `CondensedPhysics3.py` under `SESSION_302_CALCULATORS`. Smoke-test result: 20 / 20.

7. Falsifiable Predictions

(i) Galaxies with $M_{\rm BH}>1.1\times 10^{8}\,M_\odot$ host \textbf{zero} optical/X-ray TDE flares from solar-type stars; any detection falsifies the Hills cutoff.
(ii) Dwarf-galaxy IMBHs ($M_{\rm BH}\sim 10^{5}\,M_\odot$) host TDEs at $\sim 2.5\times 10^{-4}$ yr^{-1} per galaxy, a rate that LSST and Rubin will saturate within 5 years of operation.

8. Status

\textbf{[CLOSED]} \quad audit gap \#12 \quad cp4_id = 446 \quad session = 302.

9. References

Stone \& Metzger 2016, MNRAS 455, 859. Hills 1975, Nature 254, 295. Rees 1988, Nature 333, 523. UQFF_CALIBRATION_AUDIT.md.